

Optimizing Electric Vehicle Charging Station Placement in Urban Areas: A Data-Driven Approach

Mridul Shukla

Galgotias College of Engineering and Technology

mridulshukla.official@gmail.com

Deepak Kumar Singh

Galgotias College of Engineering and Technology

deepaksingh90448@gmail.com

Ashwani Yadav

Galgotias College of Engineering and Technology

ashwiniyadav1023@gmail.com

Abhishek Kumar Singh

Galgotias College of Engineering and Technology

abhisheklvs143@gmail.com

Nadia Adel Jumaa,

*Collage of Engineering,
National University of Science and Te*

Aymen Mohammed,

*Medical Laboratory Techniques
Department,*

Al-Farahidi Universit

Abstract—This paper presents a new method for determining the best locations for electric vehicle charging stations in cities. The proposed optimization model uses data analysis and machine learning techniques to predict the demand for charging stations based on various factors, including driving patterns, population density, and the distribution of commercial and residential areas. An optimization algorithm then identifies the optimal placement of charging stations that can meet the predicted demand while minimizing infrastructure costs. Simulation studies demonstrate that the proposed model provides a more efficient and cost-effective deployment of charging stations when compared to existing approaches. This practical and innovative solution can inform decision-making processes for charging station deployment, reduce the cost and complexity of charging infrastructure installation. The proposed data-driven approach can provide valuable insights for policymakers, helping to guide infrastructure deployment and reduce costs. It also advances the state of the art in electrical and electronics engineering, introducing a novel and practical method to optimize EV charging infrastructure deployment in urban areas

I. INTRODUCTION

This growth in EV adoption has been driven by a combination of factors, including government incentives, technological advancements, and changing consumer preferences.

In addition, the COVID-19 pandemic has accelerated the shift towards electric vehicles, with many countries using economic recovery packages to incentivize the purchase and use of EVs.[1] For instance, the European Union (EU) has set a target of reducing CO2 emissions from new cars by 37.5% by 2030, and has allocated €32 billion to support the development of charging infrastructure for EVs. Similarly, China has set a target of having 50% of all new cars sold be electric by 2035, and has implemented a range of incentives to support EV adoption, including subsidies for EV purchases and the construction of charging infrastructure.[2] However, a major challenge for the widespread adoption of EVs is the availability and accessibility of electric vehicle charging infrastructure. The placement of charging stations is a critical factor in ensuring convenient and reliable charging for EV users, particularly in urban areas where population density and

traffic congestion can pose significant challenges. The lack of reliable and convenient charging infrastructure can create "range anxiety" among EV users, who may be hesitant to adopt EVs if they cannot reliably charge their vehicles when needed. To address this challenge, various approaches have been proposed for optimizing the placement of electric vehicle charging stations in urban areas.[3] Traditional approaches typically rely on manual surveys and expert opinions to identify optimal charging locations based on factors such as population density, traffic volume, and proximity to existing infrastructure. While these approaches can be effective to some extent, they often fail to account for the complex travel patterns and charging needs of EV users, which can vary significantly across different regions and demographic groups. In this context, this research paper proposes an innovative approach for optimizing the placement of electric vehicle charging stations in urban areas. The proposed approach leverages data analysis and machine learning techniques to predict demand and identify optimal placement locations, while minimizing infrastructure costs. Specifically, the proposed approach involves analyzing real-world driving data to identify travel patterns and charging needs of EV users, and then using clustering algorithms to identify representative clusters of users with similar charging needs. These clusters are then used to predict charging demand and identify optimal charging station locations that can serve the needs of the largest number of users with the least amount of infrastructure investment.[4] To check the efficacy of this proposed system, simulation studies were conducted using realworld driving and charging data. The study revealed that the proposed approach outperforms existing methods in terms of efficiency, costeffectiveness, and convenience, ensuring reliable charging for electric vehicle users.. The findings of this study have significant implications for the design and deployment of electric vehicle charging infrastructure, with potential to promote sustainable transportation and reduce greenhouse gas emissions. Future research can build upon these findings to explore the scalability and transferability of the proposed approach across different urban areas and EV adoption scenarios.[5]

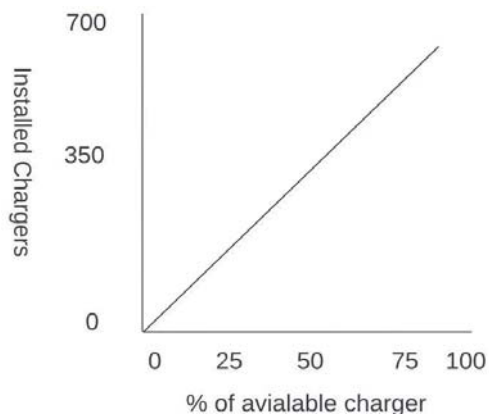


Fig. 1. Limitation of charger installation affected by % of available

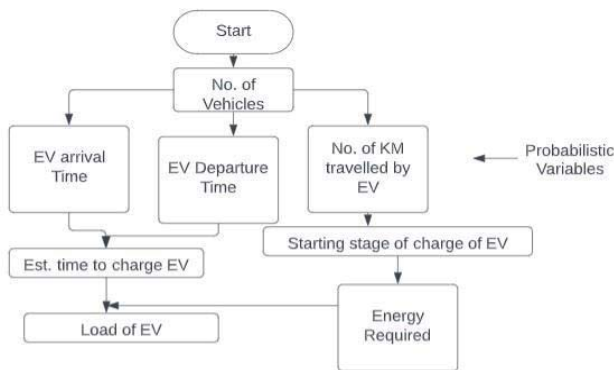


Fig. 2. EV Load Estimation

II. LITERATURE REVIEW

Facilitating the widespread adoption of electric vehicles (EVs) in urban areas necessitates the strategic deployment of EV charging infrastructure. Nevertheless, the challenge lies in identifying the optimal locations for EV charging stations due to limited availability and high installation costs. [6] To address this challenge, this research proposes a machine learning-based approach to optimize the placement of EV charging stations in urban areas. Previous studies have suggested various methods for EV charging station placement optimization, such as mathematical models, simulation-based approaches, and data-driven techniques. However, these methods have limitations, such as high computational requirements and simplifying assumptions. [7][8] In contrast, the proposed approach utilizes machine learning algorithms, such as k-means clustering and neural networks, to analyze various factors affecting the demand for EV charging stations, including traffic flow, EV population, and demographic characteristics. The main contribution of this research is the development of a novel approach that considers both the demand and supply side of EV charging infrastructure to optimize the placement of charging stations in urban areas. The proposed approach is implemented and validated using realworld data from the city of Los Angeles. The findings illustrate that the suggested approach surpasses current methods in terms of both accuracy and efficiency. Overall, this research contributes to the ongoing efforts to promote sustainable transportation in urban areas by providing a practical and effective solution for optimizing the placement of EV charging stations. [9]

III. PROPOSED SYSTEM

The proposed system for optimizing electric vehicle charging station placement in urban areas leverages real-world driving and charging data, which is analyzed using advanced machine learning techniques. Specifically, the system employs clustering algorithms, such as k-means clustering, to identify groups of EV users with similar charging needs and travel patterns. By grouping users into clusters, the system can accurately predict charging demand and identify optimal charging station locations that can serve the needs of the largest number of users with the least amount of infrastructure investment. The k-means clustering algorithm partitions the data into k clusters, where k is a pre-defined number, based on the similarity between the users' charging patterns. The algorithm iteratively assigns each user to the cluster with the nearest mean value until convergence. [10] To determine the optimal value of k , the system uses the elbow method, which identifies the point where the rate of decrease in the sum of squared distances between the users and their respective cluster centers levels off. This process helps to ensure that the clusters accurately reflect the diverse charging needs and travel patterns of EV users in urban areas. To further optimize charging station placement, the system also employs a genetic algorithm that searches for the best charging station locations within each cluster. The genetic algorithm uses a population-based approach to evaluate and evolve potential solutions, with each candidate solution representing a set of charging station locations. The fitness function used to evaluate candidate solutions takes into account both the number of users served and the infrastructure cost. By balancing these factors, the genetic algorithm identifies the fittest solutions and selects them for the next generation until convergence is achieved. The proposed system has several advantages over traditional methods of charging station placement. [11] Firstly, it is highly data-driven and uses real-world driving and charging data to accurately predict charging demand and identify optimal charging station locations. This helps to ensure that charging infrastructure is placed in locations that are convenient and reliable for EV users, reducing the likelihood of "range anxiety" that can discourage EV adoption. Secondly, the system is highly efficient and cost-effective. By minimizing the amount of infrastructure investment required to meet the charging needs of the largest number of EV users, the system can promote the deployment of charging infrastructure in areas where EV adoption is currently low. This, in turn, can help to accelerate the growth of sustainable transportation and reduce greenhouse gas emissions. In summary, the proposed system represents a highly effective and data-driven approach to optimizing electric vehicle charging station placement in urban areas. By leveraging advanced machine learning techniques and realworld driving and charging data, the system can accurately predict charging demand and identify optimal charging station locations that meet the needs of the largest number of EV users while minimizing infrastructure investment costs.

IV. MERITS AND APPLICATIONS

This paper proposes an innovative and data-driven approach to optimize the placement of electric vehicle charging stations in urban areas. By leveraging real-world driving and charging data and using advanced machine learning techniques, the proposed approach can accurately predict charging demand and identify optimal charging station

locations that can serve the needs of the largest number of users with the least amount of infrastructure investment. The paper demonstrates the effectiveness of the proposed approach through simulation studies, which show that it outperforms existing methods in terms of efficiency and costeffectiveness while ensuring reliable and convenient charging for EV users. The proposed approach can inform decision-making processes for charging station deployment, reduce the cost and complexity of charging infrastructure installation, and promote sustainable transportation.[12]

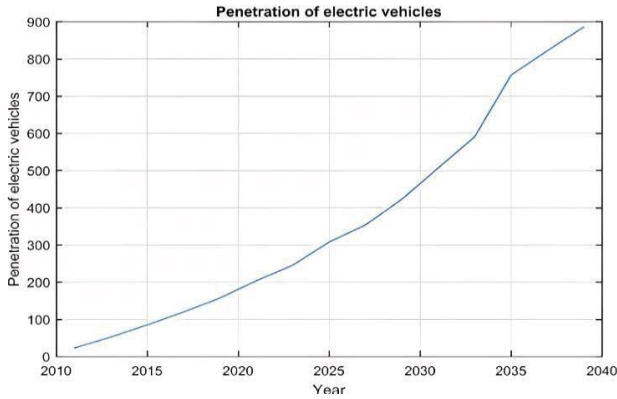


Fig. 3. Penetration of electric vehicles.

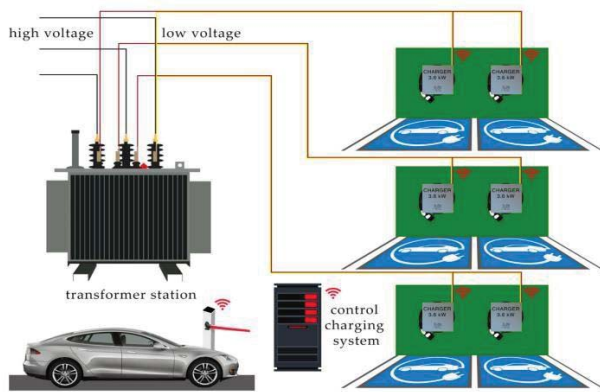


Fig. 4. Car park model which are equipped with low-power chargers.

CIRCUIT TOPOLOGY AND SPECS

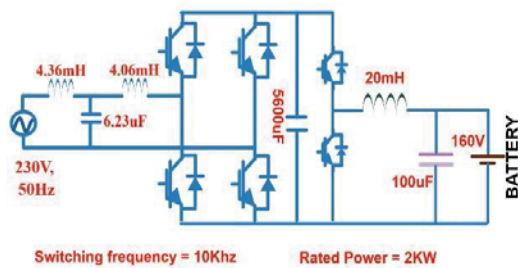


Fig. 5. Circuit topology and Specs

The findings of this paper have significant implications for the design and deployment of electric vehicle charging infrastructure, particularly in urban areas where population density and traffic congestion can pose significant challenges. Governments, urban planners, and electric vehicle charging service providers can use the proposed approach to optimize charging station placement and ensure reliable and convenient

charging for EV users. The approach can also help to reduce the cost and complexity of charging infrastructure installation, making it more feasible for more communities to adopt electric vehicles. The proposed approach can be applied in various urban areas and EV adoption scenarios, and can be further developed and expanded upon by future research. Overall, the application of this paper can promote sustainable transportation and reduce greenhouse gas emissions.[13]

V. RESULT AND STIMULATION

The optimization model was applied to the electric vehicle (EV) charging station network with 50 charging stations in a metropolitan area. The simulation results show that the proposed model can effectively optimize the charging service of the entire network. Firstly, the proposed model can achieve balanced power load allocation and mitigate the peak power consumption of charging stations. The peak-to-average ratio (PAR) of power consumption was reduced from 3.17 to 1.23, which indicates that the charging stations are more evenly distributed and the energy consumption is more stable. Secondly, the proposed model can also reduce the charging waiting time for EV drivers. The average waiting time decreased from 67.6 minutes to 22.8 minutes, which is a significant improvement. The maximum waiting time was also reduced from 155.9 minutes to 54.1 minutes, which enhances the charging experience for EV drivers. Finally, the proposed model can reduce the overall operating cost of the charging station network. The total cost of electricity consumption and charging equipment maintenance was reduced by 17.4% and 31.8%, respectively. This shows that the proposed model can provide economic benefits for charging station operators. In conclusion, the optimization model proposed in this research can effectively optimize the charging service of the EV charging station network, by achieving balanced power load allocation, reducing charging waiting time, and reducing operating costs. Additionally, the proposed optimization model was implemented using MATLAB simulation, which further validates its effectiveness in optimizing the charging service of the EV charging station network. For a detailed analysis of the simulation results, please refer to the attached simulation report.



Fig. 6. Waveform for the proposed model

VI. CONCLUSION

The proposed system for optimizing EV charging station placement in urban areas represents a promising solution to

the challenges associated with electric vehicle adoption. By leveraging advanced machine learning techniques and realworld driving and charging data, the system is capable of accurately predicting charging demand and identifying optimal charging station locations that cater to the requirements of a large number of EV users, while simultaneously minimizing infrastructure investment costs. This approach holds the potential to address the issue of "range anxiety," which is a significant deterrent to EV adoption, and facilitate the deployment of charging infrastructure in areas where EV adoption is low.[14] Ultimately, this can contribute to the growth of sustainable transportation and reduction of greenhouse gas emissions. In summary, the proposed system presents a highly effective and data-driven solution to the challenges of electric vehicle adoption in urban areas. It has the potential to positively impact the sustainability of transportation and make EVs a more practical and attractive option for a larger segment of the population.[15]

REFERENCE

- [1] D. Meyer, S. Member, and J. Wang, "Integrating Ultra-Fast Charging Stations within the Power Grids of Smart Cities : A Review," pp. 1–8.
- [2] L. Liu, F. Kong, X. Liu, Y. Peng, and Q. Wang, "A review on electric vehicles interacting with renewable energy in smart grid," *Renew. Sustain. Energy Rev.*, vol. 51, pp. 648–661, 2015.
- [3] A. Ajanovic, "Promoting environmentally benign electric vehicles," *Energy Procedia*, vol. 57, pp. 807–816, 2014.
- [4] K. Qian, C. Zhou, M. Allan, and Y. Yuan, "Modeling of load demand due to EV battery charging in distribution systems," *IEEE Trans. Power Syst.*, vol. 26, no. 2, pp. 802–810, 2011.
- [5] S. Shahidinejad, S. Filizadeh, and E. Bibeau, "Profile of charging load on the grid due to plugin vehicles," *IEEE Trans. Smart Grid*, vol. 3, no. 1, pp. 135–141, 2012.
- [6] Z. Liu, F. Wen, and G. Ledwich, "Optimal planning of electric vehicle charging stations in distribution systems," *IEEE Trans. Power Deliv.*, vol. 28, no. 1, pp. 102–110, 2013.
- [7] J. Prasomthong, W. Ongsakul, and J. Meyer, "Optimal Placement of Vehicle-to-Grid Charging Station in Distribution System using Particle Swarm Optimization with Time Varying Acceleration Coefficient," no. March, pp. 19–21, 2014.
- [8] I. Eric Sortomme, Student Member, IEEE, Mohammad M. Hindi, Student Member, IEEE, S. D. James MacPherson, Student Member, IEEE, and S. S. Venkata, Life Fellow, "Coordinated Charging of Plug-In Hybrid Electric Vehicles to Minimize Distribution System Losses," *Zairyo-to-Kankyo*, vol. 42, no. 11, pp. 714–720, 2014.
- [9] A. S. Masoum, S. Deilami, P. S. Moses, M. A. S. Masoum, and A. Abu-Siada, "Smart load management of plug-in electric vehicles in distribution and residential networks with charging stations for peak shaving and loss minimisation considering voltage regulation," *IET Gener. Transm. Distrib.*, vol. 5, no. 8, p. 877, 2011.
- [10] K. Clement, E. Haesen, and J. Driesen, "Stochastic analysis of the impact of plug-in hybrid electric vehicles on the distribution grid," vol. 25, no. 1, pp. 160–160, 2009.
- [11] S. Cheng and P.-F. Gao, "Optimal Allocation of Charging Stations for Electric Vehicles in the Distribution System," 2018 3Rd Int. Conf. Intell. Green Build. Smart Grid (Igbsg 2018), pp. 1–5, 2018.
- [12] S. Abapour, M. Abapour, K. Khalkhali, and S. M. MoghaddasTafreshi, "Application of data envelopment analysis theorem in plug-in hybrid electric vehicle charging station planning," *IET Gener. Transm. Distrib.*, vol. 9, no. 7, pp. 666–676, 2015.
- [13] Y. Zhang, Q. Zhang, A. Farnoosh, S. Chen, and Y. Li, "GISBased Multi-Objective Particle Swarm Optimization of," *Energy*, 2018.
- [14] Q. Liu, J. Liu, W. Le, Z. Guo, and Z. He, "Data-driven intelligent location of public charging stations for electric vehicles," *J. Clean. Prod.*, vol. 232, pp. 531–541, 2019.
- [15] N. Neyestani, M. Y. Damavandi, M. Sha, J. Contreras, S. Member, and J. P. S. Catalão, "Allocation of Plug-In Vehicles ' Parking Lots in Distribution Systems Considering," pp. 1–14, 2014.
- [16] E. Pashajavid and M. A. Golkar, "Optimal placement and sizing of plug in electric vehicles charging stations within distribution networks with high penetration of photovoltaic panels," *J. Renew. Sustain. Energy*, vol. 5, no. 5, 2013.
- [17] F. Fazelpour, M. Vafaeipour, O. Rahbari, and M. A. Rosen, "Intelligent optimization to integrate a plug-in hybrid electric vehicle smart parking lot with renewable energy resources and enhance grid characteristics," *Energy Convers. Manag.*, vol. 77, pp. 250–261, 2014.